

Project 2 – Automated Irrigation Controller using ESP32

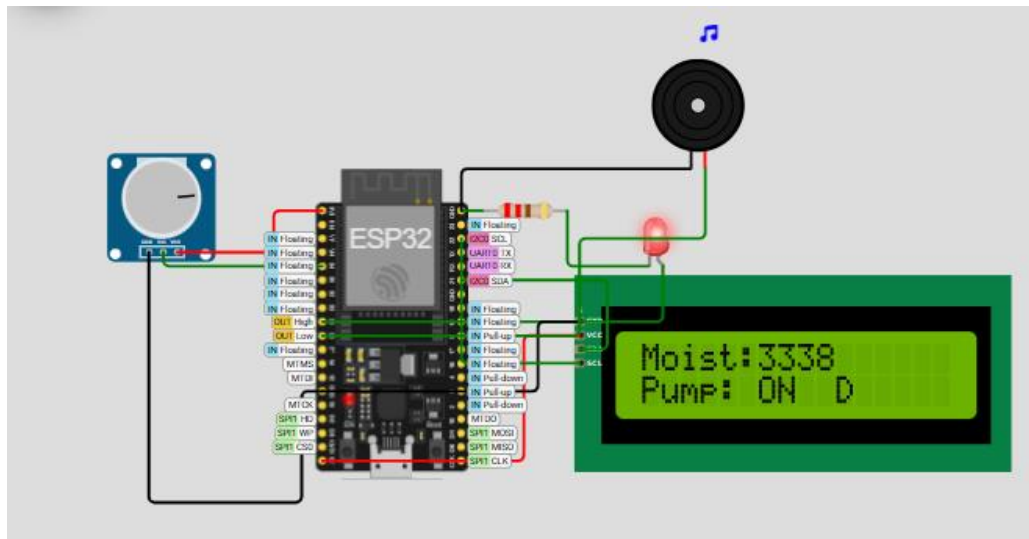
Objective:

The objective of this project is to design an automated irrigation system that monitors soil moisture and controls water supply automatically using ESP32. The system detects soil dryness and activates a pump when irrigation is needed.

Components used:

- ESP32
- Potentiometer
- LED
- 220ohm Resistor
- 16 x 2 I2C LEC Display
- Piezo Buzzer
- Jumper wires
- Wokwi simulator

Circuit diagram:



Component	ESP32 Pin
Potentiometer SIG	GPIO 34
LED(Pump)	GPIO 25
Buzzer	GPIO 26
LCD SDA	GPIO 21
LCD SCL	GPIO 22
VCC	3.3V/5V
GND	GND

Working Principle:

- Potentiometer simulates soil moisture values
- ESP32 reads analog moisture values using ADC
- Moisture level is classified as wet, normal, or dry
- Pump turns ON when soil is dry
- Buzzer alerts when soil is too dry
- LCD displays moisture value and pump status

Code:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

const int sensorPin = 34; // Potentiometer SIG
const int pumpPin = 25;   // LED (Pump)
const int buzzerPin = 26; // Buzzer

int moistureValue = 0;

// LCD Address 0x27, 16 columns, 2 rows
LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {
  Serial.begin(115200);

  pinMode(pumpPin, OUTPUT);
  pinMode(buzzerPin, OUTPUT);

  lcd.init();
  lcd.backlight();

  lcd.setCursor(0, 0);
  lcd.print(" Smart Irrig.");
  lcd.setCursor(0, 1);
  lcd.print(" Initializing");

  Serial.println("=====");
  Serial.println(" Automated Irrigation System");
  Serial.println(" DecodeLabs - Project 2");
  Serial.println("=====");

  delay(3000);
  lcd.clear();
}

void loop() {
  moistureValue = analogRead(sensorPin);
```

```

Serial.println("=====");
Serial.print("Moisture Value : ");
Serial.println(moistureValue);

lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Moist:");
lcd.print(moistureValue);

if (moistureValue > 2500) { // Dry Soil
    digitalWrite(pumpPin, HIGH);

    tone(buzzerPin, 1000);
    delay(200);
    noTone(buzzerPin);

    Serial.println("Soil Status : DRY");
    Serial.println("Pump Status : ON");
    Serial.println("Alert : BUZZER ON");

    lcd.setCursor(0, 1);
    lcd.print("Pump: ON DRY");
}

else if (moistureValue > 1000) { // Normal Soil
    digitalWrite(pumpPin, LOW);
    noTone(buzzerPin);

    Serial.println("Soil Status : NORMAL");
    Serial.println("Pump Status : OFF");

    lcd.setCursor(0, 1);
    lcd.print("Status:NORMAL");
}

else { // Wet Soil
    digitalWrite(pumpPin, LOW);
    noTone(buzzerPin);

    Serial.println("Soil Status : WET");
    Serial.println("Pump Status : OFF");

    lcd.setCursor(0, 1);
    lcd.print("Pump: OFF WET");
}

Serial.println("=====");
Serial.println();

```

```
delay(2000);}
```

Output:

```
=====
Moisture Value : 3338
Soil Status    : DRY
Pump Status    : ON
Alert          : BUZZER ON
=====
```

Result:

The automated irrigation controller was implemented successfully in Wokwi simulation and the desired output was obtained.

Conclusion:

Through this project, I learned analog sensor interfacing, ADC conversion, actuator control, LCD interfacing, and automation logic using ESP32.